

17.2 Types of sums of squares and what they test

Section 9.3 defined the Type I, II and III sums of squares by the subspaces they project onto (Definition 9.3.1, Definition 9.3.2). It also found the cell-mean hypotheses of three of them in the two-way layout: equal weighted row means for a factor entered first, equal unweighted row means for Type III, and a single column of the table for last-in sums of squares under reference coding (Theorem 9.3.2). This section completes the list with one lemma, and then shows that the hypothesis of a main effect's last-in sum of squares is decided by the coding of the *other* factor.

17.2.1 One lemma for every sum of squares

Every term of a factorial model has columns that are constant within cells, so every model matrix we meet has the form $\mathbf{W}\mathbf{A}$ for a matrix $\mathbf{A}$ with p rows, one for each cell. On the cell scale, projections become weighted by the counts: write

$$\mathbf{H}_\mathbf{A} = \mathbf{A}(\mathbf{A}^\top \mathbf{D} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{D},$$

the operator that fits a table of cell means by the columns of $\mathbf{A}$ with weights n_ω .

Lemma 17.2.1 — The hypothesis of an extra sum of squares

Let $\mathbf{X}_1 = \mathbf{W}\mathbf{A}_1$ and $\mathbf{X}_2 = \mathbf{W}\mathbf{A}_2$, let $\mathbf{P}_1$ project onto $\mathcal{C}(\mathbf{X}_1)$, and let $\mathbf{Q}$ be the projection onto $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2])$ minus $\mathbf{P}_1$, so that $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top \mathbf{Q} \mathbf{y}$. For every table of cell means $\boldsymbol{\mu}$:

- (a) $\mathbf{P}_1 \mathbf{W} \boldsymbol{\mu} = \mathbf{W} \mathbf{H}_{\mathbf{A}_1} \boldsymbol{\mu}$;
- (b) $\mathbf{Q} \mathbf{W} \boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{A}_2^\top \mathbf{D} (\mathbf{I} - \mathbf{H}_{\mathbf{A}_1}) \boldsymbol{\mu} = \mathbf{0}$;
- (c) in the cell-means model, $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1)$ is the sum of squares of the hypothesis in (b), and its F ratio with the cell-means error mean square is the F statistic Eq. 17.1.1 of that hypothesis.

Proof.

- (a) $\mathbf{X}_1^\top \mathbf{X}_1 = \mathbf{A}_1^\top \mathbf{D} \mathbf{A}_1$, so by Theorem 6.3.6 $\mathbf{P}_1 \mathbf{W} \boldsymbol{\mu} = \mathbf{W} \mathbf{A}_1 (\mathbf{A}_1^\top \mathbf{D} \mathbf{A}_1)^{-1} \mathbf{A}_1^\top \mathbf{W}^\top \mathbf{W} \boldsymbol{\mu} = \mathbf{W} \mathbf{H}_{\mathbf{A}_1} \boldsymbol{\mu}$.
- (b) By Lemma 9.2.1, $\mathbf{Q}$ projects onto $\mathcal{C}(\mathbf{T})$ with $\mathbf{T} = (\mathbf{I} - \mathbf{P}_1) \mathbf{X}_2$. A projection annihilates a vector iff the vector is orthogonal to its range, so $\mathbf{Q} \mathbf{W} \boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{T}^\top \mathbf{W} \boldsymbol{\mu} = \mathbf{0}$. By (a),

$$\mathbf{T}^\top \mathbf{W} \boldsymbol{\mu} = \mathbf{X}_2^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{W} \boldsymbol{\mu} = \mathbf{A}_2^\top \mathbf{W}^\top \mathbf{W} (\mathbf{I} - \mathbf{H}_{\mathbf{A}_1}) \boldsymbol{\mu} = \mathbf{A}_2^\top \mathbf{D} (\mathbf{I} - \mathbf{H}_{\mathbf{A}_1}) \boldsymbol{\mu}.$$

- (c) Let $\mathcal{S}_0 = \{\mathbf{W} \boldsymbol{\mu} : \mathbf{Q} \mathbf{W} \boldsymbol{\mu} = \mathbf{0}\}$, the mean vectors allowed by the hypothesis. Since $\mathcal{C}(\mathbf{Q}) \subseteq \mathcal{C}(\mathbf{W})$, the orthogonal complement of $\mathcal{S}_0$ within $\mathcal{C}(\mathbf{W})$ is $\mathcal{C}(\mathbf{Q})$, and $\mathbf{y}^\top \mathbf{Q} \mathbf{y}$ is the drop in residual sum of squares from $\mathcal{S}_0$ to $\mathcal{C}(\mathbf{W})$ (Eq. 6.5.2). That is the sum of squares of the hypothesis (Theorem 11.3.2(b)), whose F statistic is Eq. 17.1.1 by Proposition 17.1.1(d).

Part (b) says that the extra sum of squares of $\mathbf{X}_2$ tests whether the residual of the true cell means, after the terms already fitted and weighted by the counts, is orthogonal to the new term. Unless $\mathbf{D}$ is a multiple of the identity, the counts enter the hypothesis. That is the root of every difference between the types. ■

17.2.2 The complete list for two factors

Let the $a \times b$ layout have all $n_{ij} \geq 1$, with row totals $n_{i\cdot}$, column totals $n_{\cdot j}$ and $n = \sum_{ij} n_{ij}$. Besides the weighted row means $\bar{\mu}_i^w$ of Section 17.1, write

$$\bar{\mu}_{\cdot j}^w = \frac{1}{n_{\cdot j}} \sum_i n_{ij} \mu_{ij}, \quad \bar{\mu}^w = \frac{1}{n} \sum_{i,j} n_{ij} \mu_{ij}$$

for the weighted column means and the weighted grand mean. As in Chapter 9, $\text{SS}(B \mid \mu, A)$ denotes the extra sum of squares of the columns of B given the intercept and A .

Theorem 17.2.2 — The cell-mean hypotheses of the types of sums of squares

In the two-way layout with all cells occupied:

- (a) $\text{SS}(A \mid \mu)$, the Type I sum of squares of A entered first, addresses $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$;
- (b) $\text{SS}(B \mid \mu, A)$, which is both the Type I sum of squares of B entered second and the Type II sum of squares of B , addresses

$$\sum_i n_{ij} (\mu_{ij} - \bar{\mu}_i^w) = 0 \quad \text{for } j = 1, \dots, b; \quad (17.2.1)$$

- (c) $\text{SS}(A \mid \mu, B)$, the Type II sum of squares of A , addresses

$$\sum_j n_{ij} (\mu_{ij} - \bar{\mu}_{\cdot j}^w) = 0 \quad \text{for } i = 1, \dots, a; \quad (17.2.2)$$

- (d) the Type III sum of squares of A addresses $\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$;
- (e) $\text{SS}(AB \mid \mu, A, B)$, the interaction entry of every type, addresses additivity: $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl} = 0$ for all i, k, j, l .

The hypotheses have $a - 1$, $b - 1$, $a - 1$, $a - 1$ and $(a - 1)(b - 1)$ degrees of freedom, and each sum of squares is the sum of squares Eq. 17.1.1 of its hypothesis. The statements for B follow by exchanging the factors.

Proof. Parts (a) and (d) are Theorem 9.3.2(a) and (b). By Lemma 17.2.1(c), each sum of squares is the sum of squares of the hypothesis given by Lemma 17.2.1(b), and it remains to identify that hypothesis and count its degrees of freedom.

- (b) Take $\mathbf{X}_1 = [\mathbf{1}, \mathbf{Z}_A]$ and $\mathbf{X}_2 = \mathbf{Z}_B$. The columns of $\mathbf{X}_1$ span the vectors that are constant within rows, so $\mathbf{P}_1$ replaces each entry of a vector by the average of its row. Applied to $\mathbf{W}\boldsymbol{\mu}$, this gives $\bar{\mu}_i^w$ in row i . The entry j of $\mathbf{Z}_B^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{W}\boldsymbol{\mu}$ sums $\mu_{ij} - \bar{\mu}_i^w$ over the observations in column j , which is the left side of Eq. 17.2.1.

Type I, variety first				Type II				Type III			
V1	+0.35	+0.20	+0.10	V1	+0.22	+0.21	+0.13	V1	+0.22	+0.22	+0.22
V2	-0.08	-0.14	-0.11	V2	-0.13	-0.13	-0.05	V2	-0.11	-0.11	-0.11
V3	-0.05	-0.08	-0.19	V3	-0.09	-0.08	-0.08	V3	-0.11	-0.11	-0.11
	S1	S2	S3		S1	S2	S3		S1	S2	S3

Figure 17.2.1: Coefficients on the cell means of the first equation of each variety hypothesis in Example (Three varieties at three sites) (Type II divided by n_1).

Part (c) is the same argument with the factors exchanged.

For part (e), the projection is $\mathbf{Q} = \mathbf{M} - \mathbf{P}_{A+B}$, where $\mathbf{P}_{A+B}$ projects onto the additive model space, so $\mathbf{QW}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{W}\boldsymbol{\mu}$ lies in that space, that is, iff $\mu_{ij} = \alpha_i + \beta_j$ for some α, β . Additive tables have vanishing tetrads. Conversely, if all tetrads vanish, then $\mu_{ij} = \mu_{i1} + \mu_{1j} - \mu_{11}$, which is additive.

Degrees of freedom. A complete layout is connected, so the additive model matrix has rank $a + b - 1$ (Theorem 8.6.1(a)). The degrees of freedom of the extra sums of squares are therefore $a - 1$ for (a), $(a + b - 1) - a = b - 1$ for (b), $a - 1$ for (c) and $ab - (a + b - 1)$ for (e). In each case this is $\text{rank}(\mathbf{Q})$, the number of independent linear restrictions that the hypothesis places on $\boldsymbol{\mu}$.

■

In the Type II hypothesis Eq. 17.2.2, row i is compared cell by cell with the weighted means $\bar{\mu}_{\cdot j}^w$ of its columns, with the row's own counts as weights. The a equations sum to zero (Exercise (B1)). With two rows the hypothesis is $\sum_j h_j(\mu_{1j} - \mu_{2j}) = 0$ with $h_j = n_{1j}n_{2j}/n_{\cdot j}$ (Exercise (B2)), which favours columns where *both* rows are well replicated. That is the most precise comparison when the row difference is the same in every column. When it is not, the weights are set by the design rather than by anyone's question.

Under additivity, $\mu_{ij} = \alpha_i + \beta_j$, the hypotheses separate cleanly. The Type II and Type III hypotheses both reduce to $\alpha_1 = \dots = \alpha_a$: Type III because unweighted row means are $\alpha_i + \bar{\beta}$, and Type II by Theorem 9.3.2(d). The Type I hypothesis becomes

$$\alpha_i + \sum_j \frac{n_{ij}}{n_{i\cdot}} \beta_j \quad \text{the same for all } i,$$

which involves the column effects unless the counts are proportional (Theorem 9.4.3). A factor entered first is credited with the effects of the factors it is confounded with.

In Figure 17.2.1 the Type I coefficients (V1's weighted mean minus the weighted grand mean) do not sum to zero within columns, so that hypothesis compares sites as well as varieties. The Type II and Type III tables have zero column sums and compare varieties within sites. Type II gives site j the weight $n_{1j}(n_{\cdot j} - n_{1j})/n_{\cdot j}$, which is large only where V1 and the other varieties are both well replicated. Type III weights the sites equally.

17.2.3 Why Type III needs contrast codings

Definition 9.3.2 defines Type III through sum-to-zero coding, and Theorem 9.3.2(c) showed that reference coding gives a different hypothesis. The next result settles which codings give Type III and why.

Proposition 17.2.3 — Which coding a last-in sum of squares depends on

In the two-way layout with all cells occupied, code A by an $a \times (a - 1)$ coding matrix $\mathbf{C}_A$ and B by a $b \times (b - 1)$ coding matrix $\mathbf{C}_B$, with $[\mathbf{1}_a, \mathbf{C}_A]$ and $[\mathbf{1}_b, \mathbf{C}_B]$ nonsingular (Definition 8.5.1). Let the interaction columns be the products of the coded columns of A and B , and let $\mathbf{w} \neq \mathbf{0}$ span the line $\mathcal{C}(\mathbf{C}_B)^\perp \subseteq \mathbb{R}^b$. Then the partial sum of squares of the columns of A , with the intercept, B and the interaction in the model, addresses

$$\sum_j w_j \mu_{ij} \text{ is the same for all } i. \quad (17.2.3)$$

The hypothesis depends neither on $\mathbf{C}_A$ nor on the counts. It is the Type III hypothesis $\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$ iff every column of $\mathbf{C}_B$ sums to zero.

Proof. Put $\mathbf{K}_A = [\mathbf{1}_a, \mathbf{C}_A]$ and $\mathbf{K}_B = [\mathbf{1}_b, \mathbf{C}_B]$. For an observation in cell (i, j) , the product of the k th coded column of A and the l th coded column of B takes the value $(\mathbf{C}_A)_{ik}(\mathbf{C}_B)_{jl}$, the entry in row (i, j) of the corresponding column of $\mathbf{C}_A \otimes \mathbf{C}_B$. So the full model matrix is $\mathbf{W}$ times the columns of $\mathbf{K}_A \otimes \mathbf{K}_B$, namely $\mathbf{1}_a \otimes \mathbf{1}_b$, $\mathbf{1}_a \otimes \mathbf{C}_B$, $\mathbf{C}_A \otimes \mathbf{1}_b$ and $\mathbf{C}_A \otimes \mathbf{C}_B$. The matrix $\mathbf{K}_A \otimes \mathbf{K}_B$ is nonsingular (Proposition 1.10.1(b)), so the full model space is $\mathcal{C}(\mathbf{W})$ and its ab columns are linearly independent. Deleting the $a - 1$ columns $\mathbf{C}_A \otimes \mathbf{1}_b$ leaves the reduced model $\mathbf{WF}$ with

$$\mathbf{F} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{C}_B, \mathbf{C}_A \otimes \mathbf{C}_B], \quad \text{rank}(\mathbf{F}) = ab - (a - 1).$$

Let $\mathcal{V} = \{\boldsymbol{\mu} : (\mathbf{u} \otimes \mathbf{w})^\top \boldsymbol{\mu} = 0 \text{ for all } \mathbf{u} \in \mathbf{1}_a^\perp\}$. Each column of $\mathbf{F}$ lies in $\mathcal{V}$: by the mixed-product rule, $(\mathbf{u}^\top \otimes \mathbf{w}^\top)(\mathbf{1}_a \otimes \mathbf{v}) = (\mathbf{u}^\top \mathbf{1}_a)(\mathbf{w}^\top \mathbf{v}) = 0$ for any $\mathbf{v}$, and $(\mathbf{u}^\top \otimes \mathbf{w}^\top)(\mathbf{C}_A \otimes \mathbf{C}_B) = (\mathbf{u}^\top \mathbf{C}_A) \otimes (\mathbf{w}^\top \mathbf{C}_B) = \mathbf{0}$ because $\mathbf{w} \perp \mathcal{C}(\mathbf{C}_B)$. The functionals $\mathbf{u} \otimes \mathbf{w}$, $\mathbf{u} \in \mathbf{1}_a^\perp$, form a space of dimension $a - 1$, since $\mathbf{u} \mapsto \mathbf{u} \otimes \mathbf{w}$ is injective. So $\dim \mathcal{V} = ab - (a - 1) = \text{rank}(\mathbf{F})$, and $\mathcal{C}(\mathbf{F}) = \mathcal{V}$. The partial sum of squares is the drop in residual sum of squares from $\mathcal{C}(\mathbf{WF})$ to $\mathcal{C}(\mathbf{W})$, so it addresses the hypothesis $\boldsymbol{\mu} \in \mathcal{V}$. Now $(\mathbf{u} \otimes \mathbf{w})^\top \boldsymbol{\mu} = \sum_i u_i \sum_j w_j \mu_{ij}$ vanishes for every contrast $\mathbf{u}$ iff $\sum_j w_j \mu_{ij}$ does not depend on i , which is Eq. 17.2.3. Finally, $\mathbf{w} \propto \mathbf{1}_b$ iff $\mathbf{1}_b \perp \mathcal{C}(\mathbf{C}_B)$, that is, iff $\mathbf{1}_b^\top \mathbf{C}_B = \mathbf{0}$. ■

Sum-to-zero, Helmert and orthogonal polynomial codings of B all give the Type III sum of squares for A , whatever coding A receives. Reference coding of B at level r gives $\mathbf{w} = \mathbf{e}_r$ and the simple effect of Theorem 9.3.2(c). With more factors (Exercise (C2)), the last-in sum of squares of a main effect is the Type III one when every *other* factor is coded by contrasts. In statsmodels, write $\mathcal{C}(\mathbf{x}, \text{Sum})$ for every factor.

17.2.4 The variety trial, five ways

Example. (Five tables for one data set)

The table has the layout of Example (Party, education and political self-placement) in [Chapter 9](#), and the listing computes its entries the same way, as squared lengths of projections. For the trial of Example (Three varieties at three sites):

Term	df	Type I, variety first	Type I, site first	Type II	Type III	reference coding
variety	2	0.571	6.184	6.184	4.958	3.601
site	2	31.175	25.562	31.175	27.634	7.456
interaction	4	2.146	2.146	2.146	2.146	2.146
residual	28	8.781	8.781	8.781	8.781	8.781

Only the Type I columns add up to the model sum of squares 33.892; the Type II and Type III entries add up to 39.506 and 34.738. The verdicts on varieties differ. Entered first, $F = 0.91$ and $p = 0.414$: the raw averages 6.423, 6.583 and 6.275 mix each variety with the sites it was grown at. After sites (Type II), $F = 9.86$ and $p = 0.001$. Type III gives $F = 7.90$ and $p = 0.002$, the test of Example (Three varieties at three sites). The last column compares the varieties at S1 only ($F = 5.74$, $p = 0.008$).

The script checks every entry against statsmodels' `anova_lm` (types 1, 2 and 3, with sum-to-zero and Helmert coding for type 3), and recovers each entry from the matrix **L** of its part of [Theorem 17.2.2](#). It also confirms [Proposition 17.2.3](#): reference coding of the varieties with contrast coding of the sites reproduces 4.958, and a random coding of the sites gives the sum of squares of [Eq. 17.2.3](#).

```
import numpy as np
import pandas as pd
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks (any rank)."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

y = trial["y"].to_numpy()
n = len(y)
one = np.ones((n, 1))
```

```

ZA = pd.get_dummies(trial["variety"]).to_numpy(float)      # variety indicators
ZB = pd.get_dummies(trial["site"]).to_numpy(float)         # site indicators
W = np.column_stack([ZA[:, i]] * ZB for i in range(3))      # the nine cell indicators

def ss(P):
    return y @ P @ y

P0, PA, PB, PAB, M = proj(one), proj(one, ZA), proj(one, ZB), proj(one, ZA, ZB),
    proj(W)
typeI_AB = {"A": ss(PA - P0), "B": ss(PAB - PA), "AB": ss(M - PAB)} # variety
    first
typeI_BA = {"B": ss(PB - P0), "A": ss(PAB - PB), "AB": ss(M - PAB)} # site first
typeII = {"A": ss(PAB - PB), "B": ss(PAB - PA), "AB": ss(M - PAB)}

```

[Run this code online](#)

```

def coding(labels, kind):
    """Columns coding a factor: sum-to-zero, or reference level (treatment)."""
    D = pd.get_dummies(labels).to_numpy(float)
    g = D.shape[1]
    C = np.vstack([np.eye(g - 1), -np.ones(g - 1)]) if kind == "sum" else np.eye(g)
   [:, 1:]
    return D @ C

def last_in(CA, CB):
    """Partial (last-in) sums of squares of A and B with interaction columns =
    products."""
    CAB = np.column_stack([CA[:, i]] * CB for i in range(CA.shape[1]))
    full = proj(one, CA, CB, CAB)
    assert np.allclose(full, M) # every coding spans the cell-means
    space
    return {"A": ss(full - proj(one, CB, CAB)), "B": ss(full - proj(one, CA, CAB))}

typeIII = last_in(coding(trial["variety"], "sum"), coding(trial["site"], "sum"))
reference = last_in(coding(trial["variety"], "treatment"), coding(trial["site"], "
    treatment"))
sse = ss(np.eye(n) - M)
for name, t in [("Type I, variety first", typeI_AB), ("Type I, site first", typeI_BA
    ),
                ("Type II", typeII), ("Type III (sum)", typeIII), ("reference coding
    ", reference)]:
    print(f"{name:22s}", {k: round(float(v), 3) for k, v in t.items()})

```

[Run this code online](#)

The next cell builds the Type II hypothesis Eq. 17.2.2 and the Type I hypothesis as matrices on the nine cell means, and Eq. 17.1.1 recovers their entries in the table.

```

cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().to_numpy() # nine cell means, row-major
N = cells.size().to_numpy().reshape(3, 3) # counts n_ij
npi, npj = N.sum(axis=1), N.sum(axis=0)

def hypothesis_ss(L):
    """Sum of squares of  $L \mu = 0$  in the cell-means model (redundant rows allowed).
    """
    u = L @ ybar

```

```

    return u @ np.linalg.pinv(L @ np.diag(1 / N.ravel()) @ L.T) @ u

# Type II for variety: row i has coefficient n_ij (1[i = k] - n_kj / n_.j) on cell (
    k, j)
L2A = np.array([[N[i, j] * ((i == k) - N[k, j] / npj[j]) for k in range(3) for j in
    range(3)]
                for i in range(3)])
# Type I, variety first: the count-weighted variety means are equal
wrow = np.array([[ (i == k) * N[k, j] / npj[k] for k in range(3) for j in range(3)]
                for i in range(3)])
L1 = wrow[[0]] - wrow[1:]
print(f"Type II hypothesis: SS = {hypothesis_ss(L2A):.3f}; Type I hypothesis: SS =
    {hypothesis_ss(L1):.3f}")

```

[Run this code online](#)

17.2.5 Choosing a sum of squares

The advice of [Section 9.3](#) stands: decide the hypothesis first, use Type I only when the order of entry is part of the question, and without interaction prefer Type II. The cell-mean hypotheses sharpen it in three ways.

1. *Test the interaction first.* It is the same in every type and tests additivity ([Theorem 17.2.2\(e\)](#)). Under additivity Type II and Type III test the same hypothesis, and Type II is never less powerful ([Exercise \(C1\)](#)).
2. *If the factors interact,* the weights of a main-effect hypothesis decide the answer. Choose weights that describe a population (equal weights give Type III), or report simple effects within each column with multiplicity-adjusted intervals ([Chapter 13](#)).
3. *Report the hypothesis, not the type.* Anyone can read $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$, and [Eq. 17.1.1](#) tests it. With empty cells this is the only safe course ([Section 17.4](#)).

17.2.6 Exercises

17.2.6.1 A. Check your understanding

Exercise A1.

In a 2×2 layout with counts $n_{11} = 6$, $n_{12} = 2$, $n_{21} = 1$, $n_{22} = 3$, write the hypothesis of the Type I sum of squares of A entered first as one linear equation in the four cell means. Show that for the additive table $\mu_{ij} = \beta_j$, in which A has no effect, the hypothesis holds only if $\beta_1 = \beta_2$. In particular it fails for $\beta_1 = 0$, $\beta_2 = 1$.

17.2.6.2 B. Practice

Exercise B1.

Show that the left sides of the a equations [Eq. 17.2.2](#) add up to zero for every table $\boldsymbol{\mu}$.

Show also that, for a connected design, the equations have rank exactly $a - 1$. *Hint:* count degrees of freedom.

Solution. Summing over i gives

$$\sum_{ij} n_{ij} \mu_{ij} - \sum_j n_{.j} \bar{\mu}_{.j}^w = \sum_{ij} n_{ij} \mu_{ij} - \sum_j \sum_i n_{ij} \mu_{ij} = 0.$$

So at most $a - 1$ of the equations are independent. By [Lemma 17.2.1\(b\)](#) the equations are $\mathbf{Z}_A^\top (\mathbf{I} - \mathbf{P}_B) \mathbf{W} \boldsymbol{\mu} = \mathbf{0}$, where $\mathbf{P}_B$ projects onto $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$. The number of independent restrictions they impose on $\boldsymbol{\mu}$ is the rank of the projection onto $\mathcal{C}((\mathbf{I} - \mathbf{P}_B) \mathbf{Z}_A)$, which is $\text{rank}[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B] - \text{rank}[\mathbf{1}, \mathbf{Z}_B] = (a + b - 1) - b = a - 1$ for a connected design (Theorem 8.6.1(a)).

Exercise B2.

Show that when $a = 2$, the Type II hypothesis [Eq. 17.2.2](#) for A is the single equation $\sum_j h_j (\mu_{1j} - \mu_{2j}) = 0$ with $h_j = n_{1j} n_{2j} / (n_{1j} + n_{2j})$. Show that σ^2 / h_j is the variance of $\bar{y}_{1j} - \bar{y}_{2j}$, and conclude that under additivity the Type II hypothesis compares the rows with inverse-variance weights.

Solution. The first equation is

$$\sum_j n_{1j} (\mu_{1j} - (n_{1j} \mu_{1j} + n_{2j} \mu_{2j}) / n_{.j}) = \sum_j (n_{1j} n_{2j} / n_{.j}) (\mu_{1j} - \mu_{2j}),$$

and the second is its negative ([Exercise \(B1\)](#)). The variance of $\bar{y}_{1j} - \bar{y}_{2j}$ is $\sigma^2(1/n_{1j} + 1/n_{2j}) = \sigma^2 n_{.j} / (n_{1j} n_{2j}) = \sigma^2 / h_j$. If the difference $\delta = \mu_{1j} - \mu_{2j}$ is the same in every column, the most precise combination of the column differences weights them in proportion to h_j , and the hypothesis says that this combination is zero.

Exercise B3.

Suppose the counts are proportional, $n_{ij} = n_{i.} n_{.j} / n$. Show that the hypotheses of [Theorem 17.2.2\(a\)](#) and (c) coincide, and that both say that the row means weighted by the column shares $n_{.j} / n$ are equal. Why does this agree with Theorem 9.4.3?

Solution. With proportional counts, $n_{ij} / n_{i.} = n_{.j} / n =: \pi_j$ for every i , so $\bar{\mu}_i^w = \sum_j \pi_j \mu_{ij}$. The left side of [Eq. 17.2.2](#) is $n_{i.} \sum_j \pi_j (\mu_{ij} - \bar{\mu}_{.j}^w) = n_{i.} (\bar{\mu}_i^w - \sum_j \pi_j \bar{\mu}_{.j}^w)$, and $\sum_j \pi_j \bar{\mu}_{.j}^w = \bar{\mu}^w$. So (c) says $\bar{\mu}_i^w = \bar{\mu}^w$ for all i , which is (a). In Theorem 9.4.3 the centred factor spaces are orthogonal, so $\text{SS}(A \mid \mu, B) = \text{SS}(A \mid \mu)$ for every data set. Two sums of squares that agree for every $\mathbf{y}$ must address the same hypothesis.

Exercise B4.

Use [Proposition 17.2.3](#) to find the hypothesis of the last-in sum of squares of A when B (with $b = 3$) is coded by (a) the Helmert columns $(-1, 1, 0)^\top, (-1, -1, 2)^\top$; (b) the orthogonal polynomial columns $(-1, 0, 1)^\top, (1, -2, 1)^\top$; (c) the columns $(1, 0, 0)^\top, (0, 0, 1)^\top$, reference coding with level 2 as reference; (d) the columns $(1, 0, 0)^\top, (0, 1, 0)^\top$ plus

$\frac{1}{2}(1, 1, 1)^\top$ added to both.

17.2.6.3 C. Going deeper

Exercise C1.

Suppose the cell means are additive. Let $\mathbf{P}_{II} = \mathbf{P}_{A+B} - \mathbf{P}_B$ and let $\mathbf{P}_{III}$ be the projection of the Type III sum of squares of A . Both have rank $a-1$, and both are used with the cell-means error mean square. Show that the noncentrality parameters satisfy $\|\mathbf{P}_{II}\mathbf{W}\boldsymbol{\mu}\|^2 \geq \|\mathbf{P}_{III}\mathbf{W}\boldsymbol{\mu}\|^2$, so that the Type II test is at least as powerful. *Hint:* both are squared distances from $\mathbf{W}\boldsymbol{\mu}$ to a subspace.

Solution. Write $\mathbf{v} = \mathbf{W}\boldsymbol{\mu}$, which lies in the additive space $\mathcal{C}(\mathbf{P}_{A+B})$. Then $\mathbf{P}_{II}\mathbf{v} = \mathbf{v} - \mathbf{P}_B\mathbf{v}$, so $\|\mathbf{P}_{II}\mathbf{v}\|^2$ is the squared distance from $\mathbf{v}$ to $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$. Also $\mathbf{P}_{III} = \mathbf{M} - \mathbf{P}_R$, where $\mathbf{P}_R$ projects onto the reduced space $\mathcal{C}([\mathbf{1}, \mathbf{C}_B\text{-columns, interaction columns}])$, and $\mathbf{M}\mathbf{v} = \mathbf{v}$. So $\|\mathbf{P}_{III}\mathbf{v}\|^2$ is the squared distance from $\mathbf{v}$ to $\mathcal{C}(\mathbf{P}_R)$. This space contains $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$, and the distance to a larger subspace is smaller. The two F statistics have the same degrees of freedom, and the power of an F test increases with its noncentrality (Theorem 11.5.1).

Exercise C2.

In an $a \times b \times d$ layout, where C has d levels and all cells are occupied, code the factors by $\mathbf{C}_A, \mathbf{C}_B, \mathbf{C}_C$, and let the model contain all main effects and all two- and three-factor interactions, formed as products of coded columns. Show that the last-in sum of squares of the columns of A addresses equality over i of $\sum_{j,k} v_j w_k \mu_{ijk}$, where $\mathbf{v}$ spans $\mathcal{C}(\mathbf{C}_B)^\perp$ and $\mathbf{w}$ spans $\mathcal{C}(\mathbf{C}_C)^\perp$. When is this the unweighted hypothesis $\bar{\mu}_{1..} = \cdots = \bar{\mu}_{a..}$?